Treewidth and the Product of Tree-Independence and Tree-Chromatic Numbers

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October 13, 2026

Abstract

For a graph G, let $\mathrm{tw}(G)$, $\mathrm{tree}\text{-}\alpha(G)$, and $\mathrm{tree}\text{-}\chi(G)$ denote its treewidth, tree-independence number, and tree-chromatic number, respectively. We prove that every nonempty graph G satisfies

$$\mathrm{tw}(G)+1 \le \mathrm{tree}\text{-}\alpha(G)\,\mathrm{tree}\text{-}\chi(G)\big(1+\lfloor \log_2 \mathrm{tree}\text{-}\alpha(G)\rfloor\big).$$

This improves the quadratic bound of Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [3]. For every pair of integers $n, k \ge 2$, we construct a graph G with $\mathrm{tree}\text{-}\alpha(G) = n$, $\mathrm{tree}\text{-}\chi(G) = k$, and $\mathrm{tw}(G) = \Omega(nk \log_2 n)$. Thus the upper bound is best possible up to a constant factor.

Key Words: Treewidth, tree-independence number, tree-chromatic number

1 Introduction

All graphs in this paper are finite and simple. For a graph G, let $\alpha(G)$, $\omega(G)$, and $\chi(G)$ denote its independence number, clique number, and chromatic number, respectively. For $X \subseteq V(G)$, let $G[X]$ denote the subgraph of G induced by X.

A *tree-decomposition* of G is a pair $\mathcal{T} = (T, \{X_t\}_{t \in V(T)})$, where T is a tree and each $X_t \subseteq V(G)$ is a *bag*, such that:

1. for every edge $uv \in E(G)$, some bag X_t contains both u and v;
2. for every vertex $v \in V(G)$, the set $\{t \in V(T) : v \in X_t\}$ induces a nonempty connected subgraph of T.

The *width* of $\mathcal{T}$ is $\max_{t \in V(T)} |X_t| - 1$. The *treewidth* of G, denoted by $\mathrm{tw}(G)$, is the minimum width over all tree-decompositions of G.

Treewidth was introduced independently by Halin [2] in 1976 and by Robertson and Seymour [4], who developed it as part of the theory of graph minors.

The *independence number* of a tree-decomposition $\mathcal{T} = (T, \{X_t\}_{t \in V(T)})$ is $\alpha(\mathcal{T}) := \max_{t \in V(T)} \alpha(G[X_t])$. The minimum of $\alpha(\mathcal{T})$ over all tree-decompositions $\mathcal{T}$ of G is its *tree-independence number*, denoted by $\operatorname{tree}\text{-}\alpha(G)$, which was introduced independently by Yolov [6] and by Dallard, Milanič, and Štorgel [1].

Similarly, the *chromatic number* of $\mathcal{T}$ is $\chi(\mathcal{T}) := \max_{t \in V(T)} \chi(G[X_t])$. The minimum of $\chi(\mathcal{T})$ over all tree-decompositions $\mathcal{T}$ of G is its *tree-chromatic number*, denoted by $\operatorname{tree}\text{-}\chi(G)$, which was introduced by Seymour [5].

Motivated by the elementary inequality $|V(G)| \leq \alpha(G)\chi(G)$, Dallard, Milanič, and Štorgel [1] asked the following question.

Question 1.1 (Dallard, Milanič, and Štorgel [1, Question 8.4])**.** *Is the inequality*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G) \operatorname{tree}\text{-}\chi(G)$$

valid for every graph G?

Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [3] disproved the inequality in Question 1.1 and established the following quadratic upper bound.

Theorem 1.2 (Koutsoutis et al. [3, Proposition 3])**.** *For every graph G,*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G)^2 \operatorname{tree}\text{-}\chi(G).$$

Using Seymour's separation lemma [5, Theorem 1.1], we improve Theorem 1.2 as follows.

Theorem 1.3. *Every nonempty graph G satisfies*

$$\operatorname{tw}(G) + 1 \leq \operatorname{tree}\text{-}\alpha(G) \operatorname{tree}\text{-}\chi(G)\big(1 + \lfloor \log_2 \operatorname{tree}\text{-}\alpha(G) \rfloor\big).$$

For the lower bound, Koutsoutis et al. [3] proved the following result.

Theorem 1.4 (Koutsoutis et al. [3, Theorem 2])**.** *Let $f : \mathbb{R} \to \mathbb{R}_{>0}$ and $h : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ be functions. Suppose that $h(x)/x$ is nondecreasing and $h(x) = o(x \log\log x)$ as $x \to \infty$. Then some graph G satisfies*

$$\operatorname{tw}(G) > h\big(\operatorname{tree}\text{-}\alpha(G)\big) f\big(\operatorname{tree}\text{-}\chi(G)\big).$$

Furthermore, we obtain the following lower bound.

Theorem 1.5. *There is an absolute constant $c > 0$ such that, for all integers $n, k \geq 2$, there exists a graph $G_{n,k}$ satisfying*

$$\operatorname{tree}\text{-}\alpha(G_{n,k}) = n, \qquad \operatorname{tree}\text{-}\chi(G_{n,k}) = k, \qquad \operatorname{tw}(G_{n,k}) + 1 \geq c\, nk \log_2 n.$$

Theorem 1.5 shows that the upper bound in Theorem 1.3 is best possible up to a constant factor.

2 Proof of the upper bound

A *separation* of G is a pair (A, B) of subsets of $V(G)$ such that $A \cup B = V(G)$ and there is no edge between $A \setminus B$ and $B \setminus A$. Seymour [5, Theorem 1.1] proved the following lemma.

Lemma 2.1 (Seymour [5, Theorem 1.1])**.** *Every graph G has a separation (A, B) such that*

$$\chi(G[A \cap B]) \le \text{tree-}\chi(G)$$

and

$$\min\{\chi(G[A \setminus B]), \chi(G[B \setminus A])\} \ge \chi(G) - \text{tree-}\chi(G).$$

Using Lemma 2.1, we prove the following result.

Lemma 2.2. *Every nonempty graph H satisfies*

$$\chi(H) \le \text{tree-}\chi(H)\big(1 + \lfloor \log_2 \alpha(H) \rfloor\big).$$

Proof. We proceed by induction on $\alpha(H)$. If $\alpha(H) = 1$, then the conclusion is immediate. We may therefore assume that $\alpha(H) \ge 2$ and $\chi(H) > \text{tree-}\chi(H)$.

By Lemma 2.1, there is a separation (A, B) of H such that both $H[A \setminus B]$ and $H[B \setminus A]$ have chromatic number at least $\chi(H) - \text{tree-}\chi(H) > 0$ and hence are nonempty. Since there are no edges between these two subgraphs, we have

$$\alpha(H[A \setminus B]) + \alpha(H[B \setminus A]) \le \alpha(H).$$

One of them, say H', therefore satisfies $1 \le \alpha(H') \le \lfloor \alpha(H)/2 \rfloor$. As $\text{tree-}\chi(H') \le \text{tree-}\chi(H)$, using Lemma 2.1 again and the induction hypothesis, we obtain

$$\begin{aligned}\chi(H) &\le \text{tree-}\chi(H) + \chi(H') \le \text{tree-}\chi(H) + \text{tree-}\chi(H')\big(1 + \lfloor \log_2 \alpha(H') \rfloor\big) \\ &\le \text{tree-}\chi(H)\big(2 + \lfloor \log_2 \lfloor \alpha(H)/2 \rfloor \rfloor\big) = \text{tree-}\chi(H)\big(1 + \lfloor \log_2 \alpha(H) \rfloor\big),\end{aligned}$$

as required. □

Proof of Theorem 1.3. Choose a tree-decomposition $\mathcal{T} = (T, \{X_t\}_{t \in V(T)})$ of G with $\alpha(\mathcal{T}) = \text{tree-}\alpha(G)$. Let X_t be a bag of maximum cardinality. Then $\alpha(G[X_t]) \le \text{tree-}\alpha(G)$ and $\text{tree-}\chi(G[X_t]) \le \text{tree-}\chi(G)$. By Lemma 2.2,

$$\begin{aligned}|X_t| &\le \alpha(G[X_t])\chi(G[X_t]) \le \alpha(G[X_t])\,\text{tree-}\chi(G[X_t])\big(1 + \lfloor \log_2 \alpha(G[X_t]) \rfloor\big) \\ &\le \text{tree-}\alpha(G)\,\text{tree-}\chi(G)\big(1 + \lfloor \log_2 \text{tree-}\alpha(G) \rfloor\big).\end{aligned}$$

By the definition of treewidth, we have

$$\text{tw}(G) + 1 \le \text{tree-}\alpha(G)\,\text{tree-}\chi(G)\big(1 + \lfloor \log_2 \text{tree-}\alpha(G) \rfloor\big),$$

as desired. □

3 A matching lower bound

We use the following definition and lemma from Koutsoutis, Krause, Liu, Redzic, and Ueckerdt [3, Lemmas 5 and 6]. For a graph H, its 1-*completion* $C(H)$ is obtained by adding, for each pair of nonadjacent vertices $u, v \in V(H)$, a new vertex whose neighborhood is exactly $\{u, v\}$.

Lemma 3.1 (Koutsoutis et al. [3, Lemmas 5 and 6])**.** *Let H be a graph with at least three vertices and at least one edge. Then*

$$\text{tree-}\chi(C(H)) = \text{tree-}\chi(H), \quad \text{tree-}\alpha(C(H)) \le \alpha(H) \quad \textit{and} \ \operatorname{tw}(C(H)) + 1 = |V(H)|.$$

For a graph G and a positive integer q, the *lexicographic product* $G[K_q]$ is obtained by replacing each vertex $v \in V(G)$ by a clique K_v of order q. For distinct $u, v \in V(G)$, K_u is complete to K_v if and only if $uv \in E(G)$, and anticomplete to K_v otherwise. For a positive integer r, the join $G \vee K_r$ is obtained by adding a clique K_r of order r and making it complete to G; for $r = 0$, this means G itself.

Lemma 3.2. *Let G be a nonempty graph with* $\text{tree-}\chi(G) = \omega(G)$*, and let $q \ge 1$ and $r \ge 0$ be integers. If $H = G[K_q] \vee K_r$, then*

$$\text{tree-}\alpha(H) = \text{tree-}\alpha(G), \quad \text{tree-}\chi(H) = q\,\text{tree-}\chi(G) + r \quad \textit{and} \ \operatorname{tw}(H)+1 = q\big(\operatorname{tw}(G)+1\big)+r.$$

Proof. We first prove the first equality. To show $\text{tree-}\alpha(H) \le \text{tree-}\alpha(G)$, choose a tree-decomposition $(T_G, \{U_t\}_{t \in V(T_G)})$ of G with nonempty bags such that $\alpha(G[U_t]) \le \text{tree-}\alpha(G)$ for every $t \in V(T_G)$. Set $T_H := T_G$. For each $t \in V(T_G)$, put $V_t := V(K_r) \cup \bigcup_{v \in U_t} V(K_v)$. Then $(T_H, \{V_t\}_{t \in V(T_H)})$ is a tree-decomposition of H. Since $\alpha(G[U_t]) = \alpha(H[V_t])$ for every $t \in V(T_G)$, we have

$$\text{tree-}\alpha(H) \le \max_{t \in V(T_H)} \alpha(H[V_t]) = \max_{t \in V(T_G)} \alpha(G[U_t]) = \text{tree-}\alpha(G).$$

Since G is isomorphic to an induced subgraph of H, we have $\text{tree-}\alpha(H) \ge \text{tree-}\alpha(G)$. Combining this with the preceding inequality gives $\text{tree-}\alpha(H) = \text{tree-}\alpha(G)$.

We next prove the second equality. To show $\text{tree-}\chi(H) \le q\,\text{tree-}\chi(G) + r$, choose a tree-decomposition $(T'_G, \{W_t\}_{t \in V(T'_G)})$ of G such that $\chi(G[W_t]) \le \text{tree-}\chi(G)$ for every $t \in V(T'_G)$. Set $T'_H := T'_G$. For each $t \in V(T'_G)$, put $X_t := V(K_r) \cup \bigcup_{v \in W_t} V(K_v)$. $(T'_H, \{X_t\}_{t \in V(T'_H)})$ is also a tree-decomposition of H. We have

$$\text{tree-}\chi(H) \le \max_{t \in V(T'_H)} \chi(H[X_t]) \le q \max_{t \in V(T'_G)} \chi(G[W_t]) + r = q\,\text{tree-}\chi(G) + r.$$

To show $\text{tree-}\chi(H) \ge q\,\text{tree-}\chi(G) + r$, recall that every clique is contained in a bag of every tree-decomposition. Using the hypothesis $\text{tree-}\chi(G) = \omega(G)$, we obtain

$$\text{tree-}\chi(H) \ge \omega(H) = q\omega(G) + r = q\,\text{tree-}\chi(G) + r.$$

Combining the two inequalities gives $\text{tree-}\chi(H) = q\,\text{tree-}\chi(G) + r$.

Finally, we prove the last equality. To show $\mathrm{tw}(H)+1 \le q(\mathrm{tw}(G)+1)+r$, choose a tree-decomposition $(T''_G, \{Y_t\}_{t\in V(T''_G)})$ of G of width $\mathrm{tw}(G)$. Set $T''_H := T''_G$. For each $t \in V(T''_G)$, put $Z_t := V(K_r) \cup \bigcup_{v\in Y_t} V(K_v)$. Again, $(T''_H, \{Z_t\}_{t\in V(T''_H)})$ is a tree-decomposition of H. We have

$$\mathrm{tw}(H)+1 \le \max_{t\in V(T''_H)} |Z_t| = q \max_{t\in V(T''_G)} |Y_t| + r = q\big(\mathrm{tw}(G)+1\big) + r.$$

To show $\mathrm{tw}(H)+1 \ge q(\mathrm{tw}(G)+1)+r$, take a tree-decomposition $(T'''_H, \{A_t\}_{t\in V(T'''_H)})$ of H of width $\mathrm{tw}(H)$. Let T'''_G be the subgraph of T'''_H induced by $\{t \in V(T'''_H) : V(K_r) \subseteq A_t\}$. If $r=0$, then $T'''_G = T'''_H$. Otherwise, T'''_G is nonempty because K_r is a clique. Since K_r is connected, T'''_G is a nonempty subtree of T'''_H. For each $t \in V(T'''_G)$, put $B_t := \{v \in V(G) : V(K_v) \subseteq A_t\}$. $(T'''_G, \{B_t\}_{t\in V(T'''_G)})$ is clearly a tree-decomposition of G.

Choose $t_0 \in V(T'''_G)$ such that $|B_{t_0}|$ is maximum. By the definition of treewidth,

$$\mathrm{tw}(G)+1 \le \max_{t\in V(T'''_G)} |B_t| = |B_{t_0}|.$$

The bag A_{t_0} contains the pairwise disjoint vertex sets $V(K_v)$ for all $v \in B_{t_0}$ and $V(K_r)$. Since each K_v has order q, we have $q|B_{t_0}| + r \le |A_{t_0}|$. The chosen tree-decomposition of H has width $\mathrm{tw}(H)$, so $|A_{t_0}| \le \mathrm{tw}(H)+1$. Consequently,

$$q\big(\mathrm{tw}(G)+1\big) + r \le q|B_{t_0}| + r \le |A_{t_0}| \le \mathrm{tw}(H)+1.$$

Combining the two inequalities gives $\mathrm{tw}(H)+1 = q(\mathrm{tw}(G)+1)+r$. □

For convenience, we restate Theorem 1.5 here.

Theorem 1.5. *There is an absolute constant $c>0$ such that, for all integers $n,k \ge 2$, there exists a graph $G_{n,k}$ satisfying*

$$\text{tree-}\alpha(G_{n,k}) = n, \qquad \text{tree-}\chi(G_{n,k}) = k, \qquad \mathrm{tw}(G_{n,k})+1 \ge c\,nk\log_2 n.$$

Proof. Fix an integer $n \ge 3$ and put $m := \lfloor \log_3 n \rfloor$. Let

$$\mathcal{I}_n := \big\{[x, x+3^i] : i, x \in \mathbb{Z},\ 0 \le i \le m,\ 0 \le x \le n-3^i\big\}.$$

Define H_n to be the graph with vertex set $\mathcal{I}_n$ in which two distinct vertices are adjacent if and only if they can be written as $[x,y]$ and $[z,w]$ such that

$$x < z \le y < w \qquad \text{and} \qquad y - z \in \{0,1,3,9,\ldots\}. \tag{1}$$

Claim 3.1. $\alpha(H_n) \le n$.

Proof. Let $\mathcal{S}$ be an independent set of H_n. Form a graph F on $\{0,\ldots,n\}$ by including the edge xy for every $[x,y] \in \mathcal{S}$, and orient each edge from its smaller endpoint to its larger endpoint. There is no directed path $x \to y \to z$ of length two, since $[x,y]$ and $[y,z]$ would be adjacent in H_n. Hence every nonisolated vertex of F is either a source or a sink.

Suppose that C is a cycle in F. Choose the smallest sink u on C, and let $\ell < r$ be its two neighbors on C. Let v be the other neighbor of r on C. The vertex r is a source, so v is a sink, and the choice of u gives $\ell < r < u < v$. Since $[r, u]$ is a vertex of H_n, the construction of H_n implies that $u - r$ is a power of three. By (1), the vertices $[\ell, u]$ and $[r, v]$ are adjacent, a contradiction. Thus F is a forest, and $|\mathcal{S}| = |E(F)| \leq n$. □

Claim 3.2. *H_n has a tree-decomposition of chromatic number at most two.*

Proof. Let P be the path with vertices $0, \ldots, n$ in this order. For each $t \in V(P)$, put $X_t := \{[x, y] \in V(H_n) : x \leq t \leq y\}$. Clearly, $(P, \{X_t\}_{t \in V(P)})$ is a tree-decomposition of H_n.

We show that $\chi(H_n[X_t]) \leq 2$ for every $t \in V(P)$. Fix $t \in V(P)$. Give a vertex $[x, y] \in X_t$ color 1 if $(x + y)/2 < t$, and color 2 if $(x + y)/2 > t$. Since $y - x$ is odd and t is an integer, $(x + y)/2 \neq t$, so every vertex receives a color.

Let $[x, y]$ and $[z, w]$ be adjacent vertices in X_t, labelled so that $x < z \leq y < w$. Since both vertices belong to X_t, we have $z \leq t \leq y$. If $z = t = y$, then $(x + y)/2 < t < (z + w)/2$. Otherwise, $z < y$. By (1) and the definition of H_n, $y - x$, $y - z$, and $w - z$ are powers of three. Since $y - z$ is smaller than both $y - x$ and $w - z$, each interval has length at least three times $y - z$, and hence $(x + y)/2 < t < (z + w)/2$. Therefore, this coloring is proper. Thus $\chi(H_n[X_t]) \leq 2$ for every $t \in V(P)$, as required. □

Counting intervals of each length gives

$$|V(H_n)| = \sum_{i=0}^{m} (n + 1 - 3^i) = (m + 1)(n + 1) - \frac{3^{m+1} - 1}{2} = n \log_3 n + O(n). \tag{2}$$

The last equality follows from $m = \log_3 n + O(1)$ and $3^{m+1} \leq 3n$. We also have the uniform estimate

$$|V(H_n)| \geq \sum_{i=0}^{m-1} (n + 1 - 3^i) \geq \frac{2mn}{3} \geq \frac{n \log_2 n}{3 \log_2 3}, \tag{3}$$

since $3^i \leq n/3$ for $i < m$ and $\log_3 n < m + 1 \leq 2m$.

The vertices $[0, 1]$ and $[1, 2]$ are adjacent, so Claim 3.2 gives tree-$\chi(H_n) = 2$. Applying Lemma 3.1 together with Claim 3.1 yields

$$\text{tree-}\chi(C(H_n)) = 2, \qquad \text{tree-}\alpha(C(H_n)) \leq n, \qquad \text{tw}(C(H_n)) + 1 = |V(H_n)|.$$

To make the tree-independence number exactly n, let B_n be the disjoint union of $C(H_n)$ and $K_{n,n}$. Clearly, tree-$\alpha(K_{n,n}) = n$, tree-$\chi(K_{n,n}) = 2$, and $\text{tw}(K_{n,n}) = n$. The intervals of lengths one and three show that $|V(H_n)| \geq 2n - 2 \geq n + 1$. By the behavior of the parameters under disjoint union,

$$\text{tree-}\alpha(B_n) = n, \qquad \text{tree-}\chi(B_n) = \omega(B_n) = 2, \qquad \text{tw}(B_n) + 1 = |V(H_n)|.$$

For $n = 2$, set $B_2 := K_{2,2}$. Combining this case with (3), we obtain, for every integer $n \geq 2$,

$$\text{tree-}\alpha(B_n) = n, \qquad \text{tree-}\chi(B_n) = \omega(B_n) = 2, \qquad \text{tw}(B_n) + 1 \geq \frac{n \log_2 n}{3 \log_2 3}. \tag{4}$$

Now fix any integer $k \geq 2$, and put $q := \lfloor k/2 \rfloor$ and $r := k - 2q \in \{0, 1\}$. Define

$$G_{n,k} := B_n[K_q] \vee K_r.$$

Lemma 3.2 gives

$$\text{tree-}\alpha(G_{n,k}) = n, \qquad \text{tree-}\chi(G_{n,k}) = 2q + r = k.$$

Since $q \geq k/3$ for all $k \geq 2$, the same lemma and (4) give

$$\text{tw}(G_{n,k}) + 1 = q\big(\text{tw}(B_n) + 1\big) + r \geq \frac{nk \log_2 n}{9 \log_2 3}.$$

Thus the theorem holds with the absolute constant $c = 1/(9 \log_2 3)$. □

Statement of AI Use. The central proof idea in this manuscript was developed with the assistance of GPT 6.1 Astra. The author subsequently refined the idea and wrote the manuscript, using the same model to assist with language polishing. The author takes full responsibility for the content of this paper.

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